Ben Lonnqvist

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Education

2020- Ph.D. Candidate, Neuroscience
EPFL (École polytechnique fédérale de Lausanne) | Lausanne, Switzerland
- Laboratory of Psychophysics–PI: Prof Michael Herzog
- Primary interests: neural network modelling of vision; visual crowding; experimental design
- Teaching: Teaching assistant for Real Analysis 2020, 2021

2016-2020 M.A. Hons, Economics and Finance
University of Aberdeen | Aberdeen, Scotland
- First-class Honours degree
- Quantitative focus in electives
- Relevant coursework: Econometrics, Mathematical and Statistical Methods in Economics, Stochastic processes, Understanding Statistics, Proof-based Microeconomics

Research

Publications

2021 **Lonnqvist, B.**, Bornet, A., Doerig, A., Herzog, M. H. (2021). A comparative biology approach to DNN modeling of vision: A focus on differences, not similarities. Publisher: *Journal of Vision*. DOI: https://doi.org/10.1167/jov.21.10.17

2020 **Lonnqvist, B.**, Clarke, A. D. F., Chakravarthi, R. (2019). Crowding in humans is unlike that in convolutional neural networks. Publisher: *Neural Networks*. DOI: https://doi.org/10.1016/j.neunet.2020.03.021

Pre-prints

2020 **Lonnqvist, B.**, Elsner, M., Hunt, A. R., Clarke, A. D. F. (2020). Modeling individual variation in visual search with reinforcement learning. PsyArXiv. DOI: https://doi.org/10.31234/osf.io/suj28.

Conference oral presentations

2019 **Lonnqvist, B.**, Clarke, A. D. F., Chakravarthi, R. (2019). Object Recognition in Deep Convolutional Neural Networks is Fundamentally Different to That in Humans. Abstract presented at the 18th annual meeting of the Scottish Vision Group in Scotland, UK.

Conference poster presentations

2021 **Lonnqvist, B.**, Doerig, A., Bornet, A., Francis, G., Schmittwilken, L., Herzog, M. H. (2021). How crowding challenges (feedforward) convolutional neural networks. Poster presented at the 2021 Vision Sciences Society meeting. DOI: https://doi.org/10.1167/jov.21.9.2039.

Invited talks

2019 **Lonnqvist, B.**, Clarke, A. D. F., Hunt, A. (2019). Changes in Human Eye Movement Strategies Over Time. Talk presented at a meeting of the schools of Psychology and Computer Science at the University of Essex, Britain, UK.

Research Work Experience

Mar - May 2018 - *2020*

Aberdeen School of Psychology | Undergraduate Research Assistant

- Voluntary research assistant in the Consciousness, Attention, and Perception Lab
- First-authored a paper that has now been published in *Neural Networks*
- Using Python, Keras and NumPy, programmed several architectures of Deep Neural Networks and wrote testing environments to produce novel research on visual crowding in convolutional NNs
- Used SSH tunneling to access a Linux high-power computing cluster

May - Aug 2019 - *2019*

Aberdeen School of Psychology | Research Intern

- Summer research intern at the Eye Movements and Attention Lab
- Pre-print under review by *PLOS Computational Biology*: https://psyarxiv.com/suj28/
- Using Python, modeled Bayes-optimal eye movements parameterized by human visual search data
- Using MATLAB and PsychToolbox, wrote the majority of a visual search psychophysical experiment
- Ran 65 participant-hours of eye tracking experiments using EyeLink hardware

Awards

Rank Prize Foundation Optoelectronics Studentship

2019 A £3000 studentship awarded for a summer research project.

Research visit support by the James S McDonnell Foundation Scholar Award in Human Cognition (PI: Dr A. Hunt)

2019 A £150 support for a research visit to the University of Essex in July 2019.

Skills Finland Entrepreneurship National Finalist

2012 Competed in the Finnish National Finals of the Skills competition in 2012.

Teaching

Research internship supervisor | MSc Student Lucas Brunschwig

2021- Supervision of MSc student Lucas Brunschwig at EPFL on an internship project involving the extraction and analysis of neural activations of recurrent predictive coding neural networks.

Teaching assistant | Real Analysis

2020, 2021 Teaching assistant for the Real Analysis course for mathematics undergraduates at EPFL in 2020 and 2021.

Academic service

Reviewing

Journals *Neural Networks*